

Stat 150 – Stochastic Processes

Chapter-by-chapter notes on the important ideas

Durrett, *Essentials of Stochastic Processes* (3rd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Taking notes

Find the chapter banner, then type under the section heading as you read:

`\notesec{1A}{Span and Linear Independence}`

Record big ideas with `idea`, definitions with `definition`, theorems with `theorem`, lemma, proposition, examples with `example`, and warnings with `remark` — all numbered within the chapter. Add a missing section by copying a `\notesec` slot; add a chapter with `\chapterbanner`.

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Chapter 1 Markov Chains

Durrett: transition probabilities, classification of states, stationary distributions, limit behavior, exit distributions and times

§1.1 Definitions and Examples

§1.2 Multistep Transition Probabilities

§1.3 Classification of States

§1.4 Stationary Distributions

§1.5 Detailed Balance Condition

§1.6 Limit Behavior

§1.7 Returns to a Fixed State

§1.8 Proof of the Convergence Theorem*

§1.9 Exit Distributions

§1.10 Exit Times

§1.11 Infinite State Spaces*

§1.12 Chapter Summary

Chapter 2 Poisson Processes

Durrett: the exponential distribution, defining and constructing the Poisson process, compound Poisson, thinning and superposition

§2.1 Exponential Distribution

§2.2 Defining the Poisson Process

§2.3 Compound Poisson Processes

§2.4 Transformations

§2.5 Chapter Summary

Chapter 3 Renewal Processes

Durrett: laws of large numbers, queueing theory applications, age and residual life

§3.1 Laws of Large Numbers

§3.2 Applications to Queueing Theory

§3.3 Age and Residual Life*

§3.4 Chapter Summary

Chapter 4 Continuous Time Markov Chains

Durrett: transition probabilities, limiting behavior, exit distributions and times, Markovian queues, networks

§4.1 Definitions and Examples

§4.2 Computing the Transition Probability

§4.3 Limiting Behavior

§4.4 Exit Distributions and Exit Times

§4.5 Markovian Queues

§4.6 Queueing Networks*

§4.7 Chapter Summary

Chapter 5 Martingales

Durrett: conditional expectation, examples, gambling strategies and stopping times, applications

§5.1 Conditional Expectation

§5.2 Examples

§5.3 Gambling Strategies, Stopping Times

§5.4 Applications

Chapter 6 Mathematical Finance

Durrett: the binomial model, American options, the Black–Scholes formula, calls and puts

§6.1 Two Simple Examples

§6.2 Binomial Model

§6.3 Concrete Examples

§6.4 American Options

§6.5 Black–Scholes Formula

§6.6 Calls and Puts

Appendix A Review of Probability

Durrett: probability spaces, independence, random variables and distributions, expectation, limit theorems

§A.1 Probabilities, Independence

§A.2 Random Variables, Distributions

§A.3 Expected Value, Moments

§A.4 Integration to the Limit
